KALLAX filament holders

Revision 3. 6 November 2026.

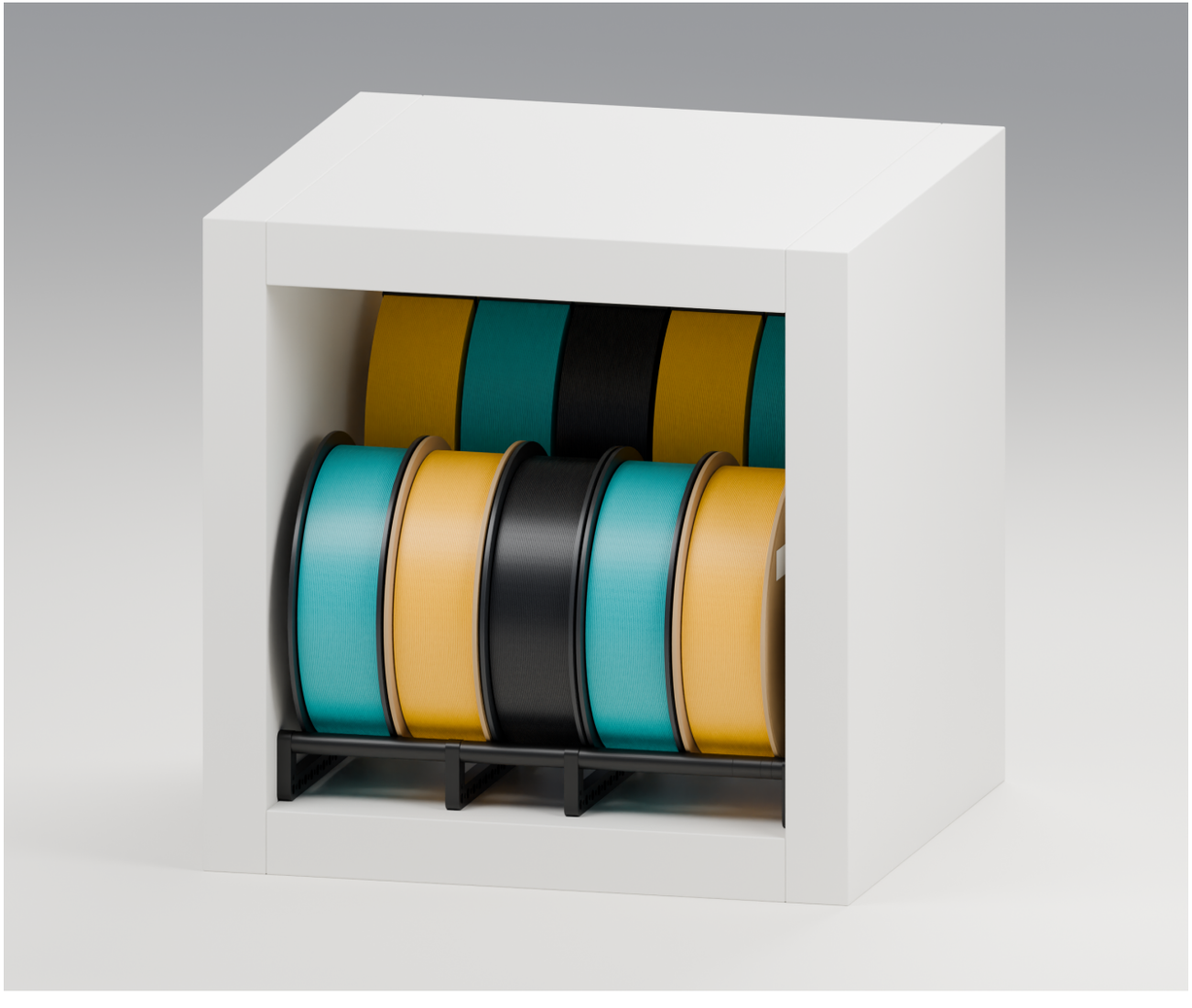

Figure 1. Five front reels. Five rear refills. Two independent holders, printed in black.

Configuration

Two independent holders per cell. Four 300 mm M6 rods per complete cell.

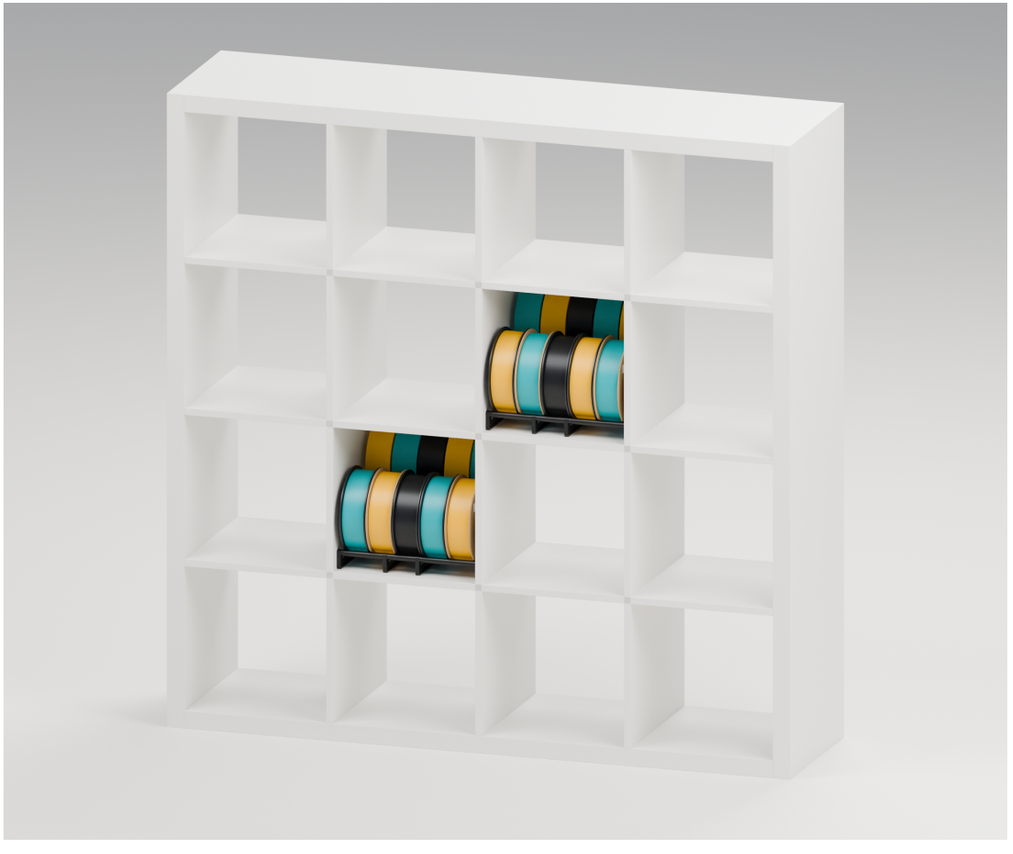

Figure 2. Two cells: four holders and eight rods.

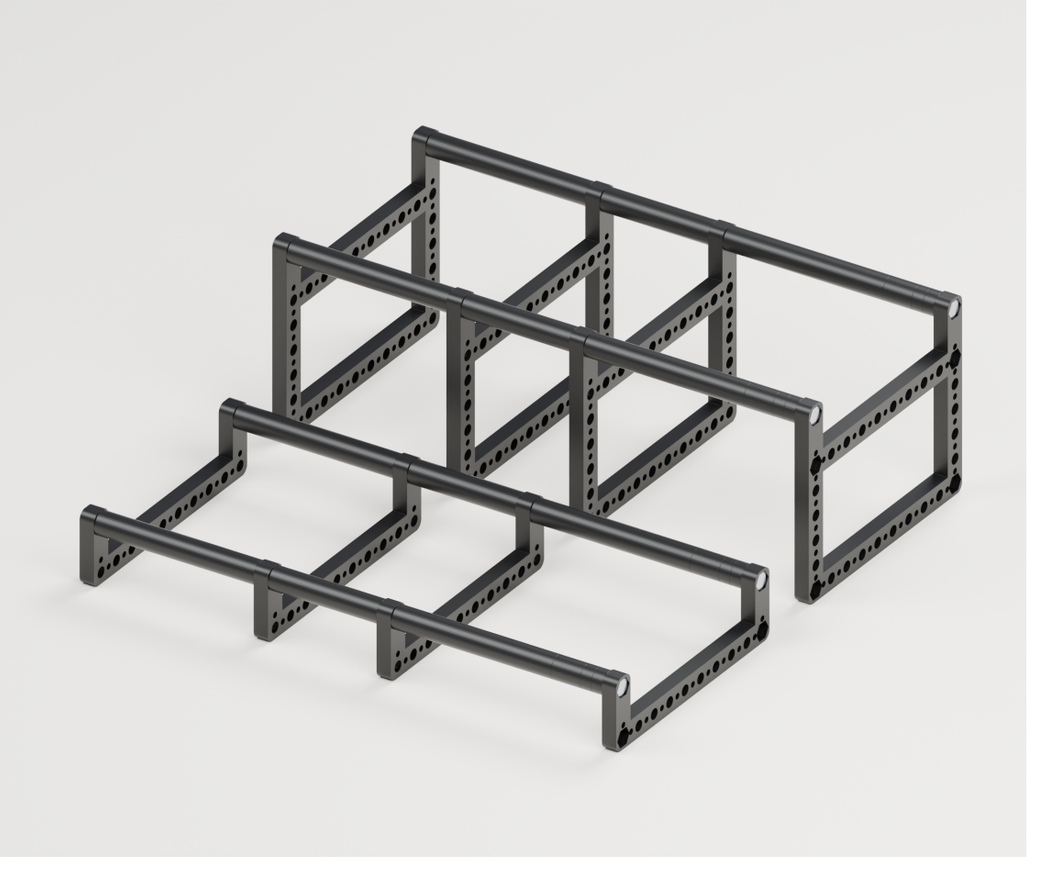

Figure 3. Four frames per holder. Short at the front; tall at the rear.

Nominal opening: 335 mm. Row centres: 187 mm. Rail heights: 37 / 112 mm.

Parts list

One complete cell: 32 printed parts and 20 purchased hardware pieces. Double all quantities for two cells.

Ref.	Type	Part	Specification	Qty.	MD5 alpha
P1	Print	Short side frame	149 x 44 x 9 mm	**4**	-
P2	Print	Tall side frame	149 x 119 x 9 mm	**4**	-
P3	Print	Long sleeve	99.67 mm long; one-end hex pocket	**8**	-
P4	Print	Centre sleeve	67.67 mm long; one-end hex pocket	**4**	-
P5	Print	Coupling housing	18 mm long; full hex bore	**4**	-
P6	Print	Retaining nut cover	7 mm long; full hex bore	**4**	-
P7	Print	Slotted cover	7 mm long; open slot	**4**	-
H1	Hardware	M6 threaded rod	300 mm cut length; M6 x 1.0	**4**	-
H2	Hardware	M6 hex nut	10 mm AF; 5 mm nominal thickness	**8**	**hcps**
H3	Hardware	M6 coupling nut	DIN 6334; 18 mm long; 10 mm AF	**4**	**1qsz8**
H4	Hardware	M6 x 25 hex-head bolt	Fully threaded; 25 mm under head	**4**	**9dg7e**

Click a part name for its STL or hardware record. MD5 alpha codes link to OOMP. H2 includes fixed and retaining nuts.

Assembly

Assemble both rails together. Repeat for the tall holder.

Figure 4. Short holder, exploded.

1. Seat the nuts

Seat H2 in the first frame. Thread in H1; keep the rod ends flush.

2. Fit the sleeves

P3 / frame / P4 / frame / P3. Repeat on both rods.

3. Fit the couplings

Insert H3 into P5. Thread onto H1 to approximately 6 mm engagement.

Adjustment

One coupling and one retained bolt per rail.

Figure 5. H3 coupling / P5 housing / P7 clip / P6 cover / H2 nut / frame / H4 bolt.

4. Fit the bolts

Face the final pockets outward. Insert H4. Add H2 loosely; slide P6 onto each bolt.

5. Set the width

Thread H4 into H3. Seat the heads. Slide P6 aside; tighten H2 against the frame; replace P6.

6. Fit the clips

Clip P7 between P5 and P6. Keep both end fasteners recessed.

Release the bolt head from its recess before turning it.

Dimensions

Axial dimensions from the fixed-end face. Millimetres.

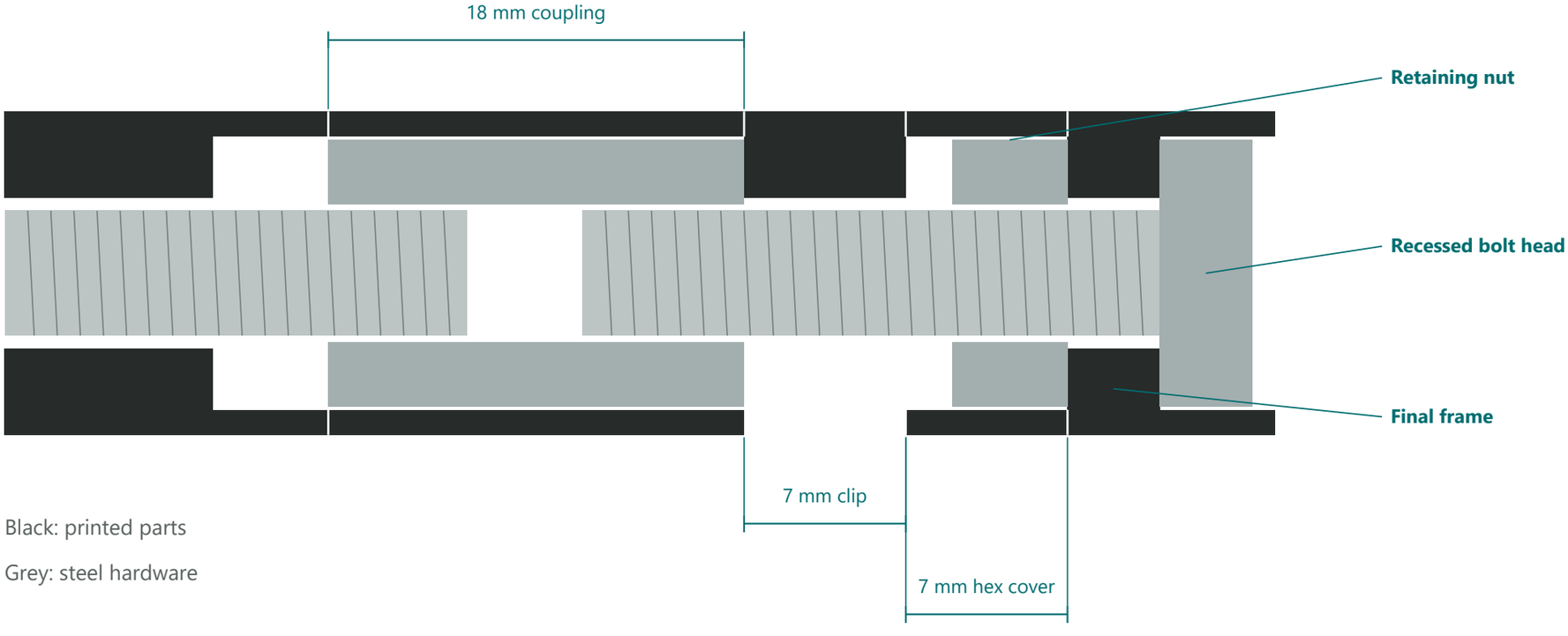

Figure 6. Retaining nut against the inner frame face; bolt head seated in the outer recess.

335.01 mm overall

4 x 9 mm frames + 267.01 mm long sleeves + 18 mm coupling housing + 7 mm slotted cover + 7 mm nut cover.

M6 x 25 mm adjustment bolt

Bearing face at 330.01 mm. Bolt tip at 305.01 mm: 7 mm nominal coupling engagement and 5.01 mm tip clearance. Rod engagement is 5.99 mm.

Nominal span: 335.01 mm. Measure the cell before setting the bolts.

Installation

Tall holder at the rear. Short holder at the front.

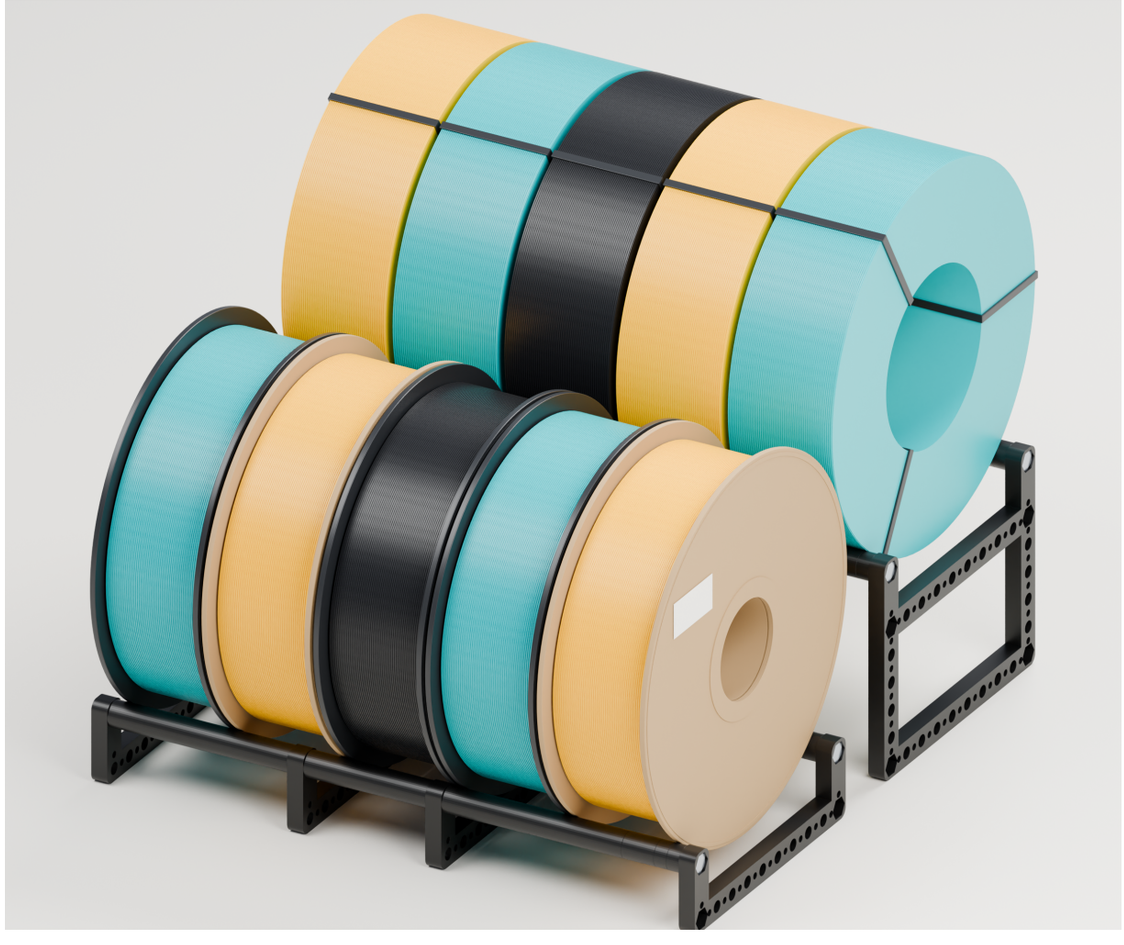

Figure 7. Generic reels: 200 mm diameter x 60 mm width. Five at 62 mm centres occupy 308 mm. Rear refills: 200 mm diameter x 58 mm width, with retaining straps fitted.

1. Set the width

Fit the holders without spreading the cabinet sides. Seat the nuts and bolt heads.

2. Position the holders

Seat all feet on the shelf. Set the row centres approximately 187 mm apart.

3. Load the rolls

Load strapped rear refills, then front reels. Leave clearance between rolls.

Part links and source files: see the repository README.